

# What NorthBay's own data says about demand, stock and where the money is

To: Head of Operations, Merchandiser, Finance lead · From: Eswar Mahalingam, Data Scientist (Zidio) · Data: 2024-01-07 → 2025-12-28 · 50 SKUs · 104 complete weeks

Units sold (2 yrs)

**510,082**

50 SKUs, 5 categories

Revenue (2 yrs)

**₹308.56 Cr**

top-10 SKUs = 48 %

Peak vs trough month

**+52 %**

March vs October

Promotion lift

**+38 %**

weekend +25 %,  
holiday -13 %

Unmapped stock

**₹17.49 Cr**

150 inventory SKUs  
with zero sales

## 1. Three things the business should act on

**1 · Demand is seasonal, not random — plan the calendar, not a flat monthly number.** Total weekly demand peaks in March and bottoms in October in *both* years, a 52 % swing. Ordering the same quantity every month guarantees spring stockouts and autumn overstock. The pattern is stable enough to forecast.

**2 · Promotions and weekends move product — and promotions are planned, so they belong in the forecast.** Promo days sell 38 % more than normal days; Saturdays/Sundays 25 % more than weekdays; national holidays sell 13 % less. A promo calendar shared with planning is worth more than any model tweak.

**3 · Money is concentrated and some of the catalogue is broken.** Ten SKUs earn 48 % of revenue; the bottom ten earn 4 % and SKU028, SKU048, SKU010, SKU050 are dead-stock candidates. **16 SKUs are priced below cost** (17 % of revenue — SKU020, SKU011, SKU038, SKU040, SKU010, SKU002...). Either the master data is wrong or NorthBay is losing money on every unit; Finance must confirm before any markdown is applied to these.



Fig 1 · Total weekly demand — the annual cycle repeats.

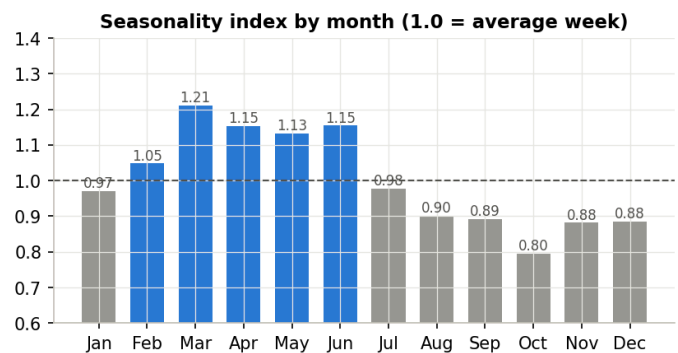


Fig 2 · Seasonality index by month (1.0 = average week).

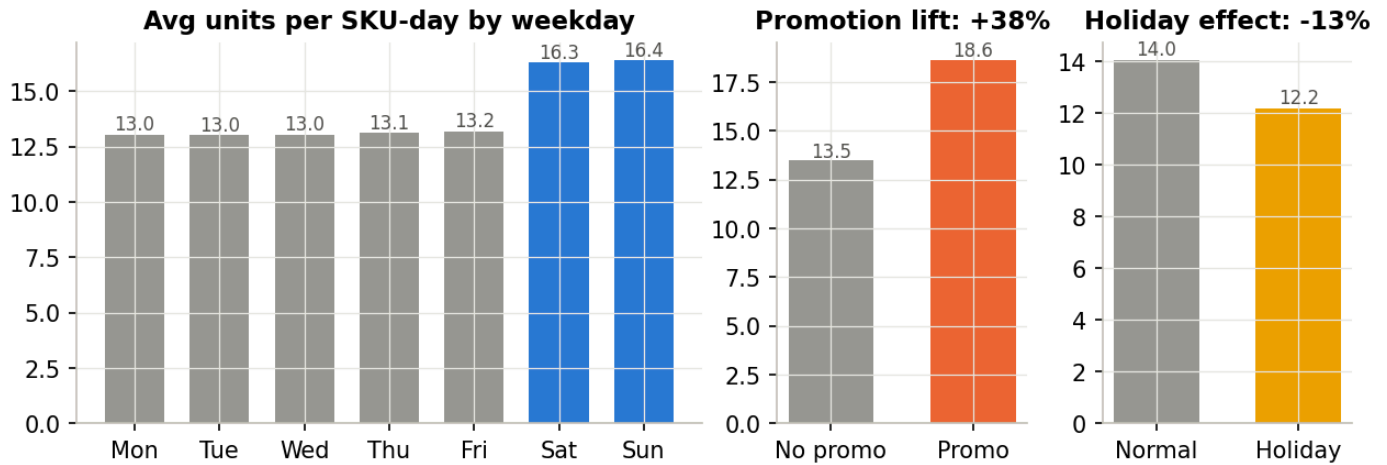


Fig 3 · Demand drivers — weekday, promotion and holiday effects (average units per SKU-day).

## 2. Top movers, dead stock and category mix

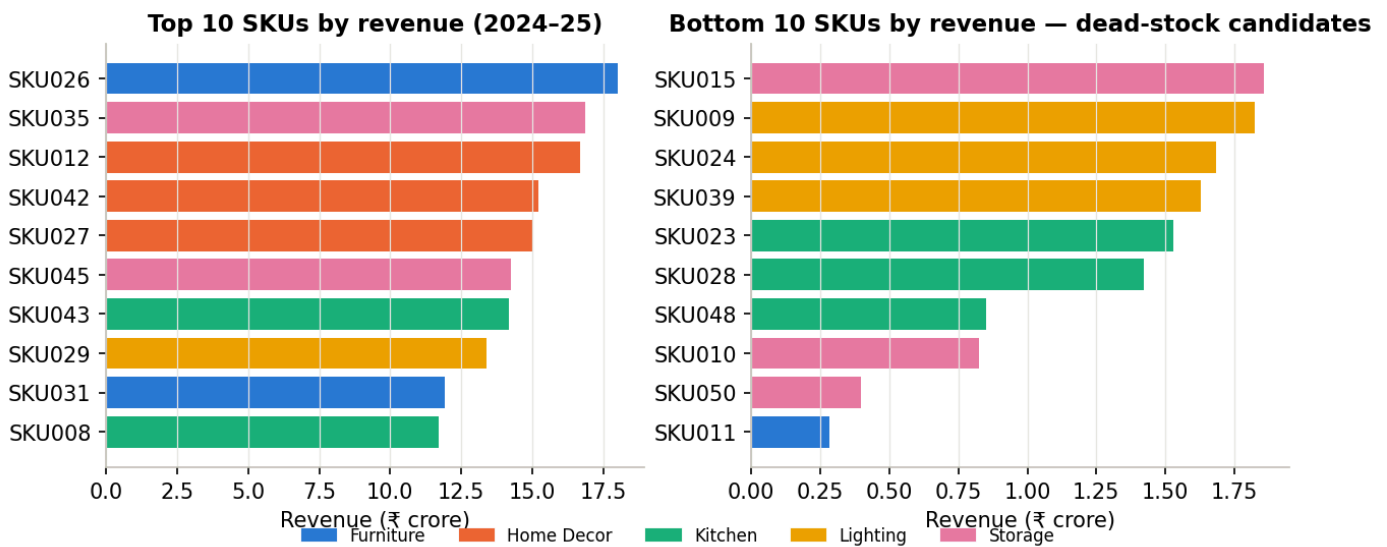


Fig 4 · Top and bottom 10 SKUs by two-year revenue, coloured by category.

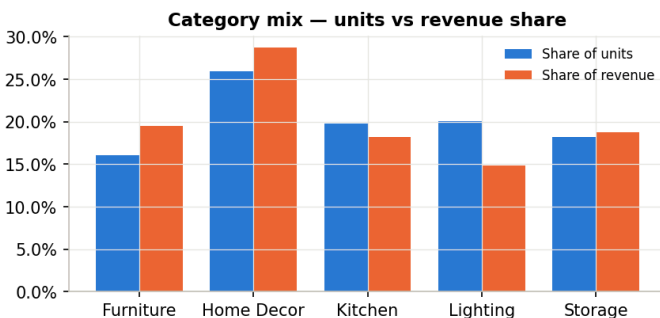


Fig 5 · Category share of units vs revenue.

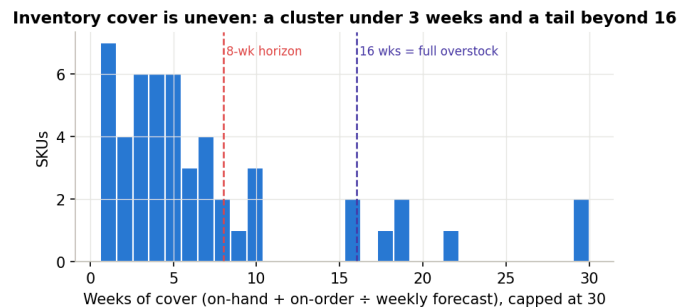


Fig 6 · Weeks of cover (position ÷ weekly forecast) is uneven: a cluster under 3 weeks and a tail beyond 16.

Each category has exactly 10 SKUs, but revenue is not evenly spread: Home Decor and Furniture carry the high-ticket items, Storage holds the most excess stock (see D4 risk scoring). The cover distribution in Fig 6 is the inventory story in one picture — NorthBay is simultaneously thin on fast sellers and deep on slow ones, which is exactly the complaint in the brief.

## 3. Data-quality issues found and how the pipeline handled them

Everything below is detected and fixed in code (`src/pipeline.py`) and written to `data/processed/data_quality_log.json` on every run; nothing is edited by hand. The sales fact table itself was clean (no duplicates, negatives or gaps) — the problems live in the dimensions.

| Table               | Issue                                                                      | n     | Handling                                                                                           | Rationale                                                                                               |
|---------------------|----------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Stock_data.csv      | Unrelated file: equity price history (AABA ticker 2006–2017)               | —     | Excluded from the pipeline                                                                         | Not part of the NorthBay star schema; no join key to any table.                                         |
| sales_daily         | Duplicate (date, sku) rows                                                 | 0     | None found — no action                                                                             | One row per SKU-day is the declared grain.                                                              |
| sales_daily         | Negative units_sold (returns?)                                             | 0     | None found — no action                                                                             | Returns are not demand.                                                                                 |
| sales_daily         | Missing numeric values                                                     | 0     | None found — no action                                                                             | A missing sales row means nothing sold that day.                                                        |
| sales_daily         | revenue != units_sold * unit_price (>₹1 gap)                               | 0     | None found — internally consistent                                                                 | Revenue is the booked figure; price is a reference.                                                     |
| sales_daily         | Rows retained after cleaning                                               | 36550 | 36550 → 36550                                                                                      |                                                                                                         |
| sku_master          | launch_date AFTER first recorded sale                                      | 21    | launch_date overridden with first_sale_date (original kept as launch_date_master)                  | Sales history is the stronger evidence; a product cannot sell before launch.                            |
| sku_master          | list_price below unit_cost (negative gross margin)                         | 16    | Kept; flagged margin_flag='negative' and surfaced to Finance                                       | Cannot be corrected from data — likely a master-data error or loss-leader; must be confirmed by client. |
| sku_master          | category/subcategory pairs that look inconsistent (e.g. Kitchen → Cushion) | 40    | Kept the master 'category' for reporting; added 'subcategory_group' as a sanity view               | Category is the client's planning hierarchy; we do not silently re-classify their catalogue.            |
| inventory_snapshots | SKUs present in inventory but with NO sales history (SKU051–SKU200)        | 150   | Excluded from forecasting/risk; listed separately as 'unsellable-or-unmapped stock' for the client | No demand signal → cannot forecast; likely discontinued or master-data gap. Raised as a finding.        |
| inventory_snapshots | Duplicate (date, sku) snapshots                                            | 0     | None found — no action                                                                             |                                                                                                         |
| inventory_snapshots | Snapshot cadence                                                           | 24    | Monthly snapshots (1st of month); latest = 2025-12-01                                              | Risk scoring uses the latest snapshot as the opening stock position at the forecast origin.             |
| calendar            | Blank holiday / promotion_event cells                                      | —     | Treated as 'None' (not missing)                                                                    | Blank means no event that day.                                                                          |
| sales_weekly        | Partial weeks at series edges (fewer than 7 days)                          | 50    | Dropped                                                                                            | A 3-day 'week' would look like a demand crash and poison the model.                                     |
| sales_weekly        | SKU-weeks with no sales rows at all                                        | 0     | Filled with 0 units                                                                                | No row = no sale.                                                                                       |

**Biggest finding for the client:** 150 SKUs in the inventory file (SKU051–SKU200) have never sold a unit in two years yet hold **₹17.49 Cr** of stock at the latest snapshot. Either they are discontinued lines that should be cleared, or the SKU codes do not match between systems. We excluded them from forecasting (no demand signal) and list them in `data/processed/inventory_unmapped_skus.csv` for Operations to reconcile.

## Analysis-ready dataset

One row per SKU per Sunday-ending week: units, revenue, promo days, holiday days, price, week-of-year, month, season, weeks since launch, category —  $50 \times 104 = 5,200$  rows, zero gaps. Forecast origin **2025-11-30** (chosen so the latest inventory snapshot and the forecast start on the same day); the 4 December weeks are a holdout the model never trains on.